GPU and TPU

Carlos Cotrini

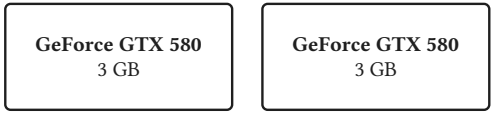
Section 1

From games to matrices

1

AlexNet, 2012

The result that started all of this was trained on two graphics cards bought for playing games.



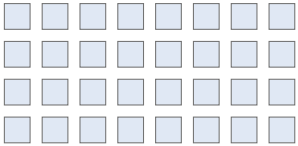
Six days. About 60 million parameters, split across the two cards because one card's memory could not hold it.

Nothing about them was designed for this

Krizhevsky, Sutskever and Hinton, *ImageNet Classification with Deep Convolutional Neural Networks*, NeurIPS 2012. No arXiv version exists.

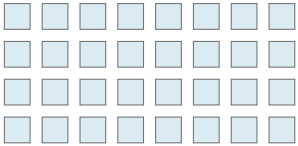
A screen and a matrix want the same thing

Shading a screen



Every pixel gets the same short recipe

Multiplying a matrix



Every output cell gets the same short recipe

- Thousands of identical, independent little jobs. That is the shape a graphics card was already built for

Section 2

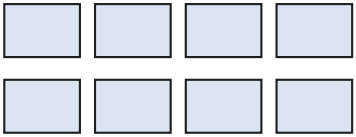
Inside the box

2

Inside the box

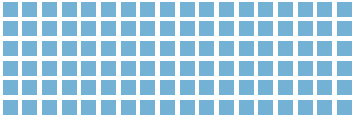
Tens of cores, or seventeen thousand

The Grace CPU



72 cores, each one clever

The Hopper GPU



16,896 cores, each one simple

235 times as many

- The CPU is built to do one hard thing quickly. The GPU is built to do one easy thing many thousands of times at once

Inside the box

They all have to do the same thing

One instruction, issued to a whole group

Give them a branch, and the ones on the wrong side wait

Half the machine idle, on work that a CPU would not blink at

- So a GPU is superb at matrices and poor at decisions, which is why the CPU on the module reads the files and runs the Python

Inside the box

The tensor cores do the matrix work

16,896 FP32 cores
the general purpose ones

528 tensor cores
they do nothing but
small matrix products

The 990 TFLOP/s you divided by at 08:00 is the tensor cores alone

Each tensor core takes two small matrices and a running total, and returns the product added to the total, in one instruction.

- A multiply and an add, which is the multiply accumulate from 08:00, done on whole tiles at a time

Inside the box

Where 990 TFLOP/s comes from

$$\underbrace{528}_{\text{tensor cores}} \times \underbrace{1024}_{\text{FLOP per cycle}} \times \underbrace{1.83\ \text{GHz}}_{\text{clock}} = 989\ \text{TFLOP/s}$$

- Peak is counted, never measured. This is the counting
- The 1024 is 512 multiply accumulates, each worth two operations

The clock here is reverse engineered from the published rate, so treat the shape as the lesson: units times clock times two.

Section 3

The other answer

3

The other answer

A TPU is built for one job

Google built its own chip because it could see how much of its datacentre was about to be doing one thing.

A GPU
a general machine that
happens to be good at this

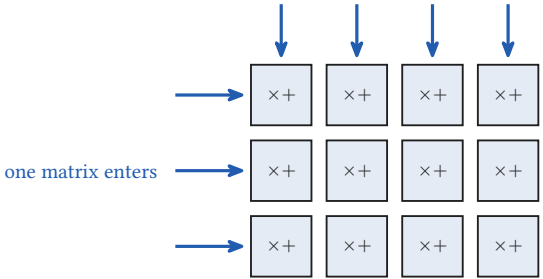
A TPU
a machine that does
this and little else

$256 \times 256 = 65{,}536$ multiply accumulate units in one grid

Jouppi et al., *In-Datacenter Performance Analysis of a Tensor Processing Unit*, ISCA 2017, arXiv:1704.04760.

The other answer

The systolic array: data walks through the grid



A value is read from memory once and then used by a whole row

- That is the answer to the 246 operations per byte from 08:00: spend less on fetching, and the arithmetic stops waiting

Section 4

What it means for you

4

What it means for you

Specialisation buys speed and costs freedom

The same trade, three times

CPU anything, slowly

GPU anything shaped like a matrix, very fast

TPU matrices, and almost nothing else, faster still

Each step down the list gives up generality and gets throughput back

- Which is why your training loop runs on all three at once: Python on the CPU, the matrices on the tensor cores

What you can draw at 11:00

Three pictures, and the third one is the number you priced this morning

- The box from 08:00 is open. A rate and a wattage, and now you know what produces them